

SECTION AAnswer **all** questions.

1. Complete the table below.

[2]

Molecule	Number of bonding pairs of electrons in outer shell	Number of lone pairs of electrons in outer shell	Shape
BeCl ₂	2	0	
PCl ₃			pyramidal

2. Complete the table below to show the type or types of bonding present in the following solids.

[2]

Solid	Type or types of bonding
calcium	
iodine	

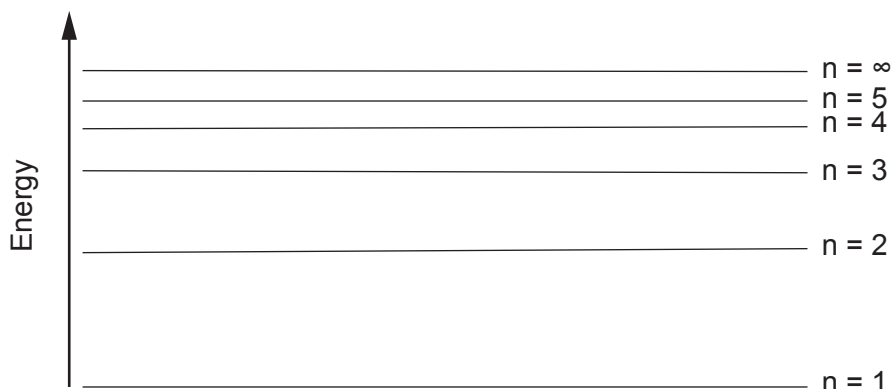
3. Give the oxidation number of rhenium in ReOCl₄.

[1]

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4. The diagram below shows the electron energy levels for a hydrogen atom.



- (a) On the diagram, draw an arrow to represent the transition corresponding to the ionisation of the atom. Label this arrow **A**. [1]
- (b) On the diagram, draw an arrow to represent the transition corresponding to the first line in the visible region in the atomic spectrum. Label this arrow **B**. [1]

5. A student said that $^{32}_{16}\text{S}^{2-}$ and $^{40}_{20}\text{Ca}^{2+}$ have the same electronic configuration.

Do you agree? Give a reason for your answer. [1]

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6. Copper can be extracted from copper(II) oxide using hydrogen.

If the atom economy for this reaction is 78% and 4.2 g of copper is formed, calculate the total mass of the reactants. [2]

Mass of reactants = g

10

